

The $p = \infty$ tree-extension endpoint is already positive

A literature-implied endpoint answer for arXiv:2301.13792

Solution packet

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Answer. Every weighted finite tree admits, at $p = \infty$, a linear extension from its leaves to all vertices with an absolute norm bound. This is a direct specialization of the universal linear Lipschitz-extension theorem for metric trees.

1 The source question

Fefferman and Klartag prove uniform linear extension for radially symmetric binary trees when $1 < p < \infty$ and leave both extreme indices open [1]. Their final formulation is reproduced below.

Remarks.

1. In this paper we have left open several natural questions, including the existence of linear extension operators for $\dot{W}^{1,p}(T)$ for weighted trees T in the extreme cases $p = 1, p = \infty$, as well as the analog of our result for the inhomogeneous Sobolev space $W^{1,p}(T)$ in place of $\dot{W}^{1,p}(T)$.
2. The problem of existence of linear Sobolev extension operators for weighted trees arose in connection with an extension problem for $W^{2,p}(\mathbb{R}^2)$. More precisely, given $E \subseteq \mathbb{R}^2$, let $\dot{W}^{2,p}(E)$ denote the space of restrictions to E of functions in $\dot{W}^{2,p}(\mathbb{R}^2)$, endowed with the natural seminorm. Does there exist a linear extension operator from $\dot{W}^{2,p}(E)$ to $\dot{W}^{2,p}(\mathbb{R}^2)$? The answer is affirmative for $p > 2$; see A. Israel [3]. For $1 < p < 2$, the answer is unknown. For a particular class of examples E , the problem reduces to the question answered by Theorem 1.1.

References

Figure 1: The exact question on page 23 of the official arXiv:2301.13792 PDF.

The $p = \infty$ question belongs to the classical theory of linear extension of scalar Lipschitz functions. It was already answered for arbitrary metric trees, without radial symmetry.

2 Literature theorem

For a metric space M , Brudnyi and Brudnyi write $\lambda(M)$ for the supremum over subsets $S \subset M$ of the least norm of a linear operator

$$E : \text{Lip}(S) \longrightarrow \text{Lip}(M)$$

which extends every scalar function on S . Their Theorem 2.4 gives, for direct p -sums of n arbitrary nontrivial metric trees and $p = 1, \infty$,

$$\lambda \left(\bigoplus_{i=1}^n \mathcal{T}_i \right) \leq cn,$$

where c is absolute [2, Theorem 2.4]. Setting $n = 1$ yields

$$\lambda(\mathcal{T}) \leq c \tag{1}$$

for every metric tree $\mathcal{T}$. The paper notes immediately afterward that the one-tree result is due to Matoušek. Its definition of a metric tree permits arbitrary positive edge lengths.

3 Deduction for weighted Sobolev traces

Let $T = (V, E)$ be any finite tree, let $L \subset V$ be its leaves, and attach a positive coefficient a_e to each edge. Define the endpoint seminorm by

$$\|F\|_{\dot{W}^{1,\infty}(V;a)} = \max_{e=uv \in E} a_e |F(u) - F(v)|. \tag{2}$$

Give edge e geometric length $\ell_e = a_e^{-1}$ and let d_T be the resulting path metric. Because a tree has a unique path between every pair of vertices, telescoping along paths shows

$$\|F\|_{\dot{W}^{1,\infty}(V;a)} = \sup_{u \neq v} \frac{|F(u) - F(v)|}{d_T(u, v)} = \|F\|_{\text{Lip}(V, d_T)}. \tag{3}$$

The trace seminorm on L equals its intrinsic Lipschitz seminorm:

$$\inf_{F|_L = f} \|F\|_{\dot{W}^{1,\infty}(V;a)} = \|f\|_{\text{Lip}(L, d_T)}. \tag{4}$$

Indeed, restriction proves “ $\geq$ ”, while the scalar McShane formula gives an extension with the same Lipschitz constant and proves “ $\leq$ ”.

Apply (1) to the geometric realization $\mathcal{T}$ of T and the subset L . There is a linear extension $E : \text{Lip}(L) \rightarrow \text{Lip}(\mathcal{T})$ with $\|E\| \leq c$. Restrict Ef to the vertices and call the resulting operator H . Equations (3)–(4) give

$$\|Hf\|_{\dot{W}^{1,\infty}(V;a)} \leq c \|f\|_{\dot{W}^{1,\infty}(L;a)}.$$

Thus H is the requested linear extension, with a constant independent of the height, branching, and edge coefficients.

If instead $p = \infty$ is defined as the literal limit of the source’s norm $(\sum_e W_e |\Delta_e F|^p)^{1/p}$, the fixed W_e disappear. The same proof applies with every $\ell_e = 1$.

4 Proof Intuition and scope

At $p = \infty$, Sobolev energy records only the largest edge slope. On a tree, controlling every edge slope is exactly the same as controlling every pairwise slope, because there is only one path to telescope along. The problem is therefore not a new weighted-energy phenomenon at this endpoint; it is the linear Lipschitz-extension problem on a metric tree, for which a universal construction was already known.

The supporting theorem predates the source question and does not identify itself as answering that question. The status is therefore *literature-implied answer*, not an explicit later resolution. This deduction answers only the $p = \infty$ part of the first source remark. It does not settle $p = 1$, the inhomogeneous problem, or the conjectured failure for arbitrary non-radial weights when $1 < p < \infty$ and $p \neq 2$.

References

- [1] C. Fefferman and B. Klartag, *Linear extension operators for Sobolev spaces on radially-symmetric binary trees*, arXiv:2301.13792, 2023.
- [2] A. Brudnyi and Yu. Brudnyi, *Metric spaces with linear extensions preserving Lipschitz condition*, arXiv:math/0404304v4, 2005.